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Ordinary Differential Equations

Higher-Order Differential Equations

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ENGR 213

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Today's Lecture

- Variation of Parameters
  - Method of variation of parameters

Reference

Advanced Engineering Mathematics, D.G. Zill (7th edition), **Section 3.5**

- Method of undetermined coefficients has an inherent weakness that limits its wider application to linear equations:
  - DE must have constant coefficients
  - g(x) must be of a certain type

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = g(x)$$

- Here, we learn a method for determining a particular solution y_p of a nonhomogeneous linear DE that has, in theory, no such restrictions on it.



Variation of Parameters Method

Method of Variation of Parameters

- To obtain a particular solution $\mathbf{y_p}$ of a linear second-order DE:

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = g(x)$$

dividing by $a_2(x)$ to obtain the standard form

$$y'' + P(x)y' + Q(x)y = f(x)$$

* Assumptions: P(x), Q(x), and f(x) are continuous on some common interval *I*.

- For the above linear second-order DE, we seek a solution of the form:

$$y_p(x) = u_1(x)y_1(x) + u_2(x)\,y_2(x)$$

* where $y_1(x)$ and $y_2(x)$ form a fundamental set of solutions on *I* of the associated homogeneous DE ($a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$).

- Next, we substitute y_p, y'_p, and y''_p in $y'' + P(x)y' + Q(x)y = f(x)$:

$$y_p'' + P(x)y_p' + Q(x)y_p = \frac{d}{dx}[y_1u_1' + y_2u_2'] + P[y_1u_1' + y_2u_2'] + y_1'u_1' + y_2'u_2'$$

$$= \frac{d}{dx}[y_1u_1' + y_2u_2'] + P[y_1u_1' + y_2u_2'] + y_1'u_1' + y_2'u_2' = f(x)$$

- Because we seek to determine two unknown functions u_1 and u_2, we need two equations: $y_1u_1' + y_2u_2' = 0$ and $y_1'u_1' + y_2'u_2' = f(x)$

- Using Cramer's rule and the Wronskian: $u_1' = \frac{W_1}{W}$ $u_2' = \frac{W_2}{W}$

  * where $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$ $W_1 = \begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix}$ $W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix}$

- Then u_1 and u_2 are obtained by integration and substituted into y_p.


QUESTIONS

EXAMPLE 1

Solve $y'' - 4y' + 4y = (x+1)\, e^{2x}$

EXAMPLE 2

Solve $4y'' + 36y = \csc 3x$

EXAMPLE 3

Solve $y'' + y = \sin x$